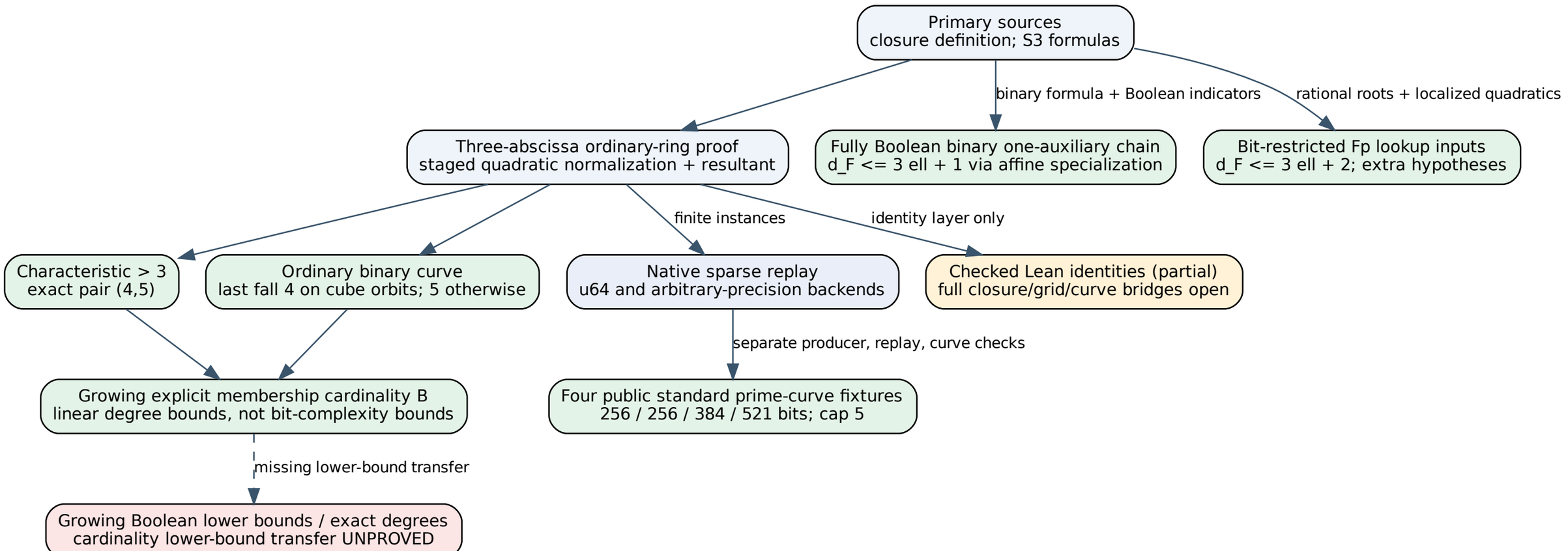

Guarded fall-degree proof dependencies (derived results)
Primary sources
closure definition; S3 formulas
binary formula + Boolean indicators
rational roots + localized quadratics
Three-abscissa ordinary-ring proof
staged quadratic normalization + resultant
Fully Boolean binary one-auxiliary chain
d_F <= 3 ell + 1 via affine specialization
Bit-restricted Fp lookup inputs
d_F <= 3 ell + 2; extra hypotheses
finite instances
identity layer only
Characteristic > 3
exact pair (4,5)
Ordinary binary curve
last fall 4 on cube orbits; 5 otherwise
Native sparse replay
u64 and arbitrary-precision backends
Checked Lean identities (partial)
full closure/grid/curve bridges open
separate producer, replay, curve checks
Growing explicit membership cardinality B
linear degree bounds, not bit-complexity bounds
Four public standard prime-curve fixtures
256 / 256 / 384 / 521 bits; cap 5
missing lower-bound transfer
Growing Boolean lower bounds / exact degrees
cardinality lower-bound transfer UNPROVED